AI Avatar Workflow

Enterprise-Ready Framework for High-Quality, Scalable Avatar Creation

Executive Summary

This comprehensive workflow enables the creation of photorealistic, brand-consistent AI avatars with full 360° fidelity and unlimited contextual variations. Designed for agencies, brands, and creative studios requiring production-grade deliverables at scale.

Metric	Standard Workflow	Enhanced Workflow
Total Time	8-10 minutes	15-25 minutes
Output Quality	Good	Production-Grade
Consistency Score	~85%	95-99%
Scalability	Limited	Enterprise-Ready
Use Cases	Basic	Comprehensive

🛠️ Tool Stack & Infrastructure

Primary Generation Tools

Category	Recommended Tools	Use Case
Base Generation	Midjourney V6, DALL-E 3, Stable Diffusion XL	Initial model creation
Consistency Engine	Weavy, IP-Adapter, InstantID	Identity preservation
Upscaling	Magnific AI, Topaz Gigapixel, Real-ESRGAN	Resolution enhancement
Face Refinement	FaceChain, PhotoMaker, GFPGAN	Facial detail optimization
Pose Control	ControlNet, DensePose, OpenPose	Precise pose manipulation

Category	Recommended Tools	Use Case
Background/Context	Banana Pro, Scenario.gg, Leonardo AI	Environment generation
Quality Assurance	Pixlr, Photoshop (Neural Filters), Lightroom	Final polish

Hardware Requirements

```
Minimum Specifications:
├── GPU: NVIDIA RTX 3080 (12GB VRAM) or equivalent
├── RAM: 32GB DDR4
├── Storage: 500GB NVMe SSD
└── Internet: 100+ Mbps (for cloud-based tools)

Recommended Specifications:
├── GPU: NVIDIA RTX 4090 (24GB VRAM)
├── RAM: 64GB DDR5
├── Storage: 2TB NVMe SSD
└── Internet: 500+ Mbps dedicated
```

📋 PHASE 1: Strategic Planning & Brand Alignment

Objective

Establish clear parameters, brand guidelines, and quality benchmarks before generation begins.

1.1 Client Discovery Questionnaire

```
## Avatar Brief Template

### Brand Identity
- Brand name: _________________
- Industry vertical: _________________
- Target demographic: _________________
- Brand personality (3-5 keywords): _________________
- Competitor references (avoid): _________________
```

```
### Avatar Specifications
- Purpose: [ ] E-commerce  [ ] Social Media  [ ] Marketing  [
] Corporate  [ ] Entertainment
- Realism level: [ ] Photorealistic  [ ] Stylized  [ ] Illust
rated
- Exclusivity: [ ] Exclusive rights  [ ] Non-exclusive  [ ] L
imited use

### Technical Requirements
- Primary platform(s): _______________
- Resolution requirements: _______________
- File format preferences: _______________
- Color profile: [ ] sRGB  [ ] Adobe RGB  [ ] P3
```

1.2 Avatar Persona Development

Attribute	Options	Considerations
Gender Expression	Male, Female, Non-binary, Androgynous	Market research, brand alignment
Age Presentation	18-25, 26-35, 36-45, 46-55, 55+	Target audience demographics
Ethnicity/Heritage	Diverse options	Market representation, cultural sensitivity
Body Type	Various builds	Inclusivity, product requirements
Style Archetype	High Fashion, Commercial, Editorial, Lifestyle, Athletic, Corporate	Brand positioning

1.3 Mood Board Creation

Required Elements:

- 5-10 reference images for overall aesthetic
- 3-5 facial structure references
- 3-5 styling/fashion references

- 2-3 lighting/atmosphere references
- Color palette extraction (primary, secondary, accent)

Output: Comprehensive creative brief document (PDF/Notion)

📋 PHASE 2: Base Model Generation

Objective

Generate multiple high-fidelity base models with optimal characteristics for consistency scaling.

2.1 Prompt Engineering Framework

Master Prompt Structure

```
[STYLE MODIFIER] [SUBJECT DESCRIPTION] [PHYSICAL ATTRIBUTES]
[CLOTHING/STYLING] [LIGHTING SETUP] [CAMERA SETTINGS]
[QUALITY MODIFIERS] [NEGATIVE PROMPTS]
```

Component Breakdown

Style Modifiers:

```
Professional fashion photography, editorial portrait,
commercial beauty shot, high-end advertising campaign,
luxury brand aesthetic, vogue magazine style
```

Physical Attribute Template:

```
[ethnicity] [gender] model, [age] years old, [face shape] face,
[eye color] eyes, [hair color] [hair length] [hair texture] hair,
[skin tone] skin with [skin quality], [distinctive features],
[body type] build, [height impression]
```

Quality Modifiers (Essential):

```
8K UHD, hyperrealistic, professional color grading,
sharp focus on face, subtle skin texture visible,
catch lights in eyes, natural skin pores,
fine hair detail, professional retouching aesthetic,
shot on Hasselblad H6D-400C, 85mm f/1.4 lens
```

Negative Prompts (Critical):

```
--no deformed, distorted, disfigured, poorly drawn,
bad anatomy, wrong anatomy, extra limbs, missing limbs,
floating limbs, disconnected limbs, mutation, mutated,
ugly, disgusting, amputation, duplicate, watermark,
signature, text, blurry, out of focus, bad quality,
low resolution, plastic skin, waxy skin, oversaturated
```

2.2 Generation Protocol

Step-by-Step Process:

1. Batch Generation (24 variations)
    - Run 4 batches of 6 images each
    - Vary seed values systematically
    - Maintain core prompt consistency
    - Document all parameters
2. First-Pass Filtering (→12 images)
    - Eliminate anatomical errors
    - Remove inconsistent lighting
    - Filter out poor compositions
    - Discard low-resolution outputs
3. Consistency Analysis (→6 images)

- Compare facial geometry
- Assess reproducibility potential
- Evaluate feature distinctiveness
- Check for "generatable" qualities

4. Client Presentation (→3 images)
    - Prepare comparison grid
    - Include rationale for each selection
    - Provide mock-up applications
5. Final Selection (→1 image)
    - Client approval
    - Document selection reasoning
    - Archive alternatives for future use

2.3 Quality Benchmarks

Criteria	Minimum Standard	Production Grade
Resolution	1024×1024	4096×4096+
Facial Symmetry	90%	95%+
Skin Texture	Visible pores	Micro-detail visible
Eye Quality	Clear irises	Catch lights + depth
Hair Detail	Grouped strands	Individual strands
Lighting	Even exposure	Dimensional + motivated

Output: 1 selected base model + 2 approved alternatives

📋 PHASE 3: Identity Lock & Model Sheet Creation

Objective

Establish immutable identity parameters and generate comprehensive reference documentation for perfect consistency.

3.1 Identity Embedding Process

Face Encoding Protocol

```
Identity Lock Workflow:
├── Extract facial landmarks (68-point detection)
├── Generate face embedding vector
├── Create identity reference pack
│   ├── Neutral expression (primary)
│   ├── Slight smile
│   ├── Profile left
│   ├── Profile right
│   └── Three-quarter views
├── Train LoRA/DreamBooth (if using SD)
└── Validate identity preservation score
```

Consistency Parameters to Lock

Parameter	Lock Method	Variance Allowed
Facial geometry	Embedding + ControlNet	<2%
Skin tone	RGB values + relative luminance	±5%
Eye color	Hex code reference	±3%
Hair color	Multiple lighting samples	±8%
Body proportions	Height ratios documented	±3%
Distinctive features	Explicit prompting	0%

3.2 Comprehensive Model Sheet

Required Views (12-Point System)

Additional Required Elements

View #	Description	Purpose	Priority
⑨	Face Close-up	Facial detail reference	Critical
⑩	Hand Detail	Gesture/product holding	High
⑪	Full Body (Standing)	Proportion reference	Critical
⑫	Full Body (Seated)	Alternative pose base	Medium

3.3 Expression Library

Minimum Expression Set (8 expressions):

```
1. Neutral (baseline)
2. Subtle smile (commercial default)
3. Genuine smile with teeth
4. Serious/Confident
5. Contemplative/Thoughtful
6. Surprised/Delighted
```

```
7. Candid laugh
8. Professional/Corporate
```

Extended Expression Set (16 expressions):
Add: Curious, Determined, Relaxed, Playful, Mysterious, Warm, Energetic, Sophisticated

3.4 Documentation Package

```
## Model Sheet Documentation

### Identity Card
- Model ID: [UNIQUE_ID]
- Creation Date: [DATE]
- Version: [X.X]
- Primary Artist/Creator: [NAME]

### Physical Specifications
| Attribute | Value | Hex/RGB |
|-----------|-------|---------|
| Skin Tone | [Description] | #[XXXXXX] |
| Eye Color | [Description] | #[XXXXXX] |
| Hair Color | [Description] | #[XXXXXX] |
| Lip Color | [Description] | #[XXXXXX] |

### Prompt DNA
[Core prompt components that produce this model]

### Consistency Score
- Self-similarity test: [XX]%
- Cross-generation stability: [XX]%
```

Output: Complete model sheet package (12+ reference images + documentation)

📋 PHASE 4: Wardrobe & Styling System

Objective

Create a comprehensive wardrobe library that maintains model consistency while enabling unlimited outfit variations.

4.1 Wardrobe Categories

https://visualize.graphy.app/view/0d793ff1-ae72-4d36-b7c7-08c55627dbf8

4.2 Styling Prompt Templates

Casual/Lifestyle

```
[MODEL REFERENCE] wearing [casual outfit description],
relaxed fit, contemporary streetwear aesthetic,
[color palette], [fabric texture visible],
styled with [accessories], [footwear]
```

Business/Corporate

```
[MODEL REFERENCE] in professional attire,
[suit/blazer/dress description], tailored fit,
executive styling, [neutral/power colors],
polished appearance, [appropriate accessories]
```

Editorial/High Fashion

```
[MODEL REFERENCE] in avant-garde fashion,
[designer aesthetic], architectural silhouettes,
editorial lighting, high contrast styling,
[bold color or monochromatic], statement pieces
```

4.3 Accessory Integration

Category	Items	Consistency Notes
Eyewear	Sunglasses, prescription frames	Match face shape
Jewelry	Watches, necklaces, earrings, rings	Consistent style tier
Bags	Handbags, backpacks, clutches	Scale appropriately
Headwear	Hats, caps, headbands	Maintain hair visibility
Footwear	Sneakers, heels, boots, sandals	Proportion consistency

Output: Wardrobe library (30-40 outfit combinations documented)

📋 PHASE 5: Context & Scenario Generation

Objective

Generate production-ready contextual images across diverse environments while maintaining perfect model consistency.

5.1 Environment Categories

Studio Environments

```
Category: Controlled Studio
├── White cyclorama (e-commerce standard)
├── Colored seamless backgrounds
├── Textured backdrops (concrete, fabric, gradient)
├── Product photography setup
└── Editorial studio with props
```

Location Environments

```
Category: Real-World Locations
├── Urban
│   ├── City streets
│   ├── Coffee shops/Cafés
│   ├── Modern architecture
│   └── Rooftop settings
├── Nature
│   ├── Beach/Coastal
│   ├── Forest/Mountains
│   ├── Parks/Gardens
```

```
│   └── Desert/Arid landscapes
├── Interior
│   ├── Luxury home
│   ├── Modern office
│   ├── Hotel/Resort
│   └── Retail environment
└── Travel
    ├── Airport/Transit
    ├── Tourist destinations
    ├── Cultural landmarks
    └── Adventure settings
```

5.2 Lighting Scenarios

https://visualize.graphy.app/view/dd8ae332-5b1c-44ea-89fc-f9a611d8590c

5.3 Pose Library

Standing Poses

Pose ID	Name	Best For	Difficulty
ST-01	Confident Stand	Corporate, Editorial	Low
ST-02	Weight Shift	Lifestyle, Fashion	Low
ST-03	Hand on Hip	Commercial, E-commerce	Low
ST-04	Arms Crossed	Corporate, Fitness	Medium
ST-05	Walking Motion	Lifestyle, Street	Medium
ST-06	Dynamic Movement	Editorial, Athletic	High

Seated Poses

Pose ID	Name	Best For	Difficulty
SE-01	Professional Seated	Corporate, Interview	Low
SE-02	Casual Lounge	Lifestyle, Home	Low
SE-03	Cross-legged	Wellness, Casual	Medium

Pose ID	Name	Best For	Difficulty
SE-04	Perched/Edge	Editorial, Modern	Medium

Action Poses

Pose ID	Name	Best For	Difficulty
AC-01	Product Interaction	E-commerce, Tech	Medium
AC-02	Device Usage	Tech, Lifestyle	Medium
AC-03	Athletic Movement	Fitness, Sport	High
AC-04	Social Interaction	Lifestyle, Group	High

5.4 Context Generation Workflow

```
For each scenario:
1. Load model reference (from Phase 3)
2. Select environment template
3. Choose lighting preset
4. Define pose requirements
5. Generate 4-6 variations
6. Quality filter (2-3 best)
7. Enhancement pass
8. Final selection
```

5.5 Scenario Prompt Structure

```
[MODEL REFERENCE/EMBEDDING], [pose description],
in [environment description], [time of day],
[lighting conditions], [atmosphere/mood],
[camera angle], [focal length equivalent],
[depth of field], [color grading style],
professional photography, magazine quality,
[technical specifications]
```

Example - Lifestyle Coffee Shop:

```
[Sarah_Avatar_v2.1], sitting at a marble café table,
holding a ceramic latte cup, genuine warm smile,
in a modern minimalist coffee shop, afternoon,
soft natural window light from the left, warm cozy atmosphere,
```

```
eye-level angle, 50mm equivalent,
shallow depth of field with bokeh background,
warm neutral color grading, lifestyle photography,
shot on Sony A7R V, f/2.0, natural skin tones
```

Output: 15-25 contextual images across 5-8 scenario categories

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📋 PHASE 6: Post-Processing & Enhancement

Objective

Elevate generated images to commercial-grade quality through systematic enhancement protocols.

6.1 Enhancement Pipeline

6.2 Upscaling Specifications

Source Resolution	Target Resolution	Upscaler	Settings
1024×1024	2048×2048	Real-ESRGAN	x2, denoise: 0.3
1024×1024	4096×4096	Magnific AI	x4, creativity: low
2048×2048	8192×8192	Topaz Gigapixel	x4, face recovery: on

6.3 Color Grading Presets

```
Preset Library:
├── Commercial Neutral (sRGB optimized)
├── Warm Lifestyle (golden undertones)
├── Cool Editorial (blue shadows)
├── High Contrast Fashion (crushed blacks)
├── Soft Luxury (lifted blacks, muted)
├── Vibrant Pop (saturated, punchy)
├── Film Emulation (Portra 400, Ektar 100)
└── Custom Brand LUT (client-specific)
```

6.4 Common Artifact Fixes

Issue	Detection	Solution
Hand Deformities	Manual review	Inpaint + ControlNet redraw
Eye Asymmetry	AI detection	Targeted face restoration
Hair Artifacts	Edge detection	Selective inpainting
Background Bleeding	Mask analysis	Clean edge refinement
Clothing Distortion	Pattern analysis	Regional regeneration
Skin Texture Loss	Detail comparison	Texture overlay/restoration

Output: Production-ready images (per specifications)

📋 PHASE 7: Quality Assurance & Delivery

7.1 Comprehensive QA Checklist

Technical Quality

- ☐ Resolution meets or exceeds specifications
- ☐ No visible compression artifacts
- ☐ Proper color space (sRGB/Adobe RGB as specified)
- ☐ Correct aspect ratios for all deliverables
- ☐ File sizes optimized without quality loss

Identity Consistency

- ☐ Facial features match reference (95%+ similarity)
- ☐ Skin tone consistent across all images
- ☐ Eye color consistent
- ☐ Hair color/style consistent
- ☐ Body proportions maintained
- ☐ Distinctive features preserved

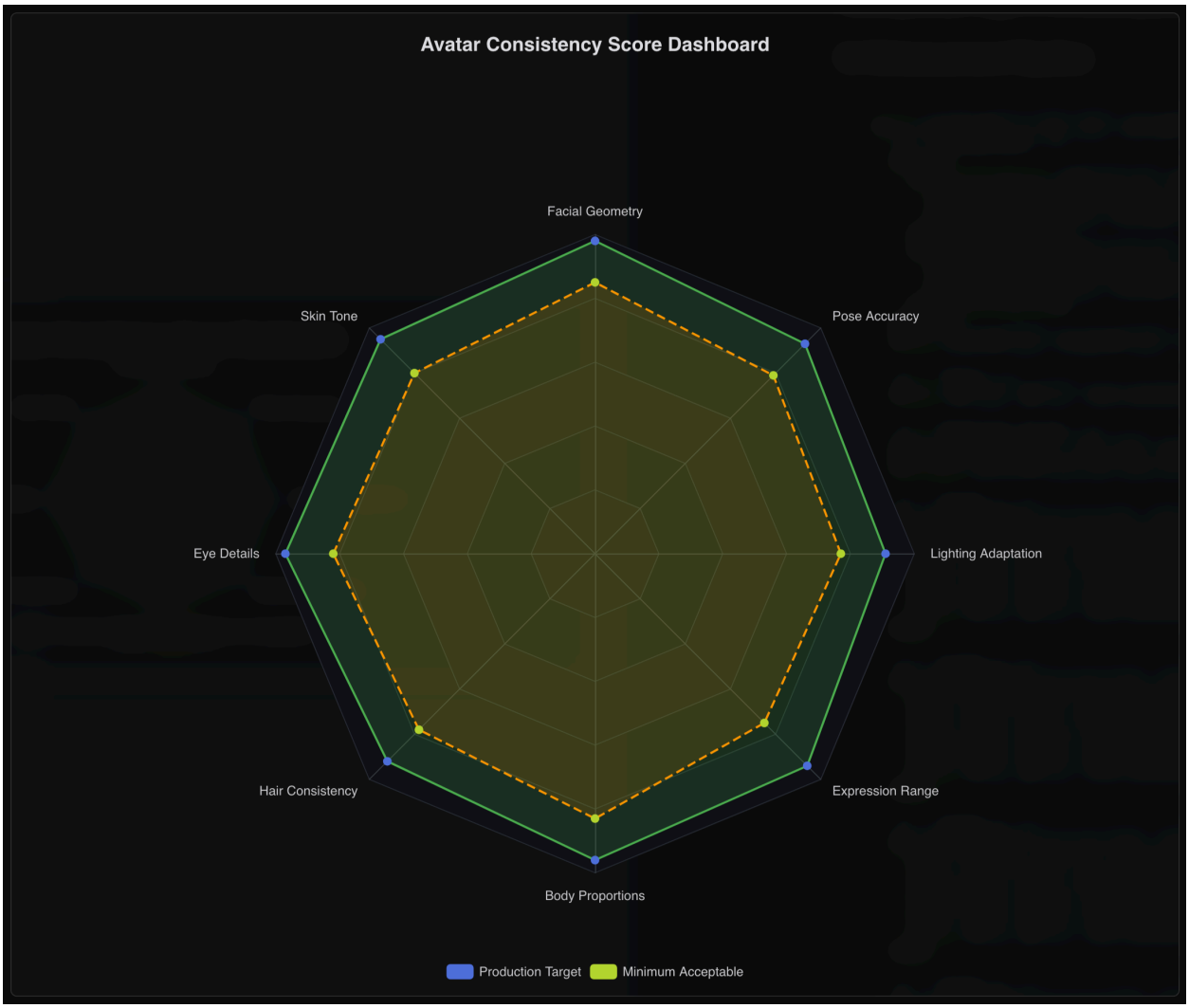
Technical Execution

- [ ] No anatomical errors
- [ ] Hands rendered correctly
- [ ] Eyes have proper catch lights
- [ ] Hair detail is natural
- [ ] Clothing fits naturally
- [ ] Backgrounds are clean/appropriate

Artistic Quality

- [ ] Lighting is motivated and consistent
- [ ] Color grading matches brief
- [ ] Composition follows guidelines
- [ ] Poses appear natural
- [ ] Expressions are appropriate
- [ ] Overall aesthetic matches brand

7.2 Consistency Scoring System



7.3 File Organization Structure

```
Project_[ClientName]_[AvatarName]_[Date]/
├── 00_Documentation/
│   ├── Creative_Brief.pdf
│   ├── Avatar_Identity_Card.pdf
│   ├── Prompt_Documentation.md
│   └── Revision_History.md
│
├── 01_Model_Sheet/
│   ├── Primary_Views/
│   │   ├── [Avatar]_Front_4K.png
│   │   ├── [Avatar]_Left_Profile_4K.png
│   │   ├── [Avatar]_Right_Profile_4K.png
│   │   └── [Avatar]_Back_4K.png
│   ├── Secondary_Views/
│   │   └── [8 three-quarter views]
```

```
│   └── Close_Ups/
│       ├── [Avatar]_Face_Detail.png
│       └── [Avatar]_Full_Body.png
│
├── 02_Expression_Library/
│   ├── [Avatar]_Neutral.png
│   ├── [Avatar]_Smile.png
│   └── [Additional expressions]
│
├── 03_Wardrobe/
│   ├── Casual/
│   ├── Business/
│   ├── Formal/
│   ├── Athletic/
│   └── Seasonal/
│
├── 04_Contextual_Images/
│   ├── Studio/
│   ├── Urban/
│   ├── Nature/
│   ├── Interior/
│   └── Lifestyle/
│
├── 05_Web_Optimized/
│   ├── Social_Media/
│   │   ├── Instagram_Square/
│   │   ├── Instagram_Story/
│   │   ├── LinkedIn/
│   │   └── Twitter/
│   └── Website/
│       ├── Hero_Images/
│       └── Thumbnails/
│
└── 06_Source_Files/
    ├── PSD_Files/
    ├── Raw_Generations/
    └── Project_Files/
```

7.4 Delivery Specifications

Use Case	Format	Resolution	Color Space	Compression
Print (High-End)	TIFF	300 DPI, 8K+	Adobe RGB	Lossless
Print (Standard)	PNG	300 DPI, 4K	sRGB	Lossless
Web (Hero)	WebP/PNG	2048px wide	sRGB	85% quality
Web (Standard)	WebP/JPEG	1200px wide	sRGB	80% quality

Use Case	Format	Resolution	Color Space	Compression
Social Media	JPEG/PNG	Platform-specific	sRGB	90% quality
Thumbnail	WebP	400px wide	sRGB	75% quality

Output: Complete delivery package with all specifications met

---

📊 Workflow Metrics & Time Allocation

Detailed Time Breakdown

https://visualize.graphy.app/view/0fecf4b0-f9c0-431b-a51d-118a21744f63

Scalability Metrics

Volume	Time per Avatar	Hourly Output	Daily Capacity
Single	25-35 min	2 avatars	16 avatars
Batch (5)	20-25 min avg	2.5 avatars	20 avatars
Batch (10)	18-22 min avg	3 avatars	24 avatars
Enterprise (20+)	15-18 min avg	3.5 avatars	28 avatars

🚀 Advanced Techniques & Pro Tips

Consistency Maximization

1. Seed Locking: Document and reuse successful seeds
2. Prompt Anchoring: Keep 80% of prompt constant, vary only context
3. Reference Weighting: Use higher reference strength (0.7-0.85) for critical features
4. Multi-stage Generation: Generate base → refine face → add context (separate passes)
5. A/B Testing: Always generate comparison sets to identify drift

Quality Acceleration

1. Template Libraries: Pre-build prompt templates for common scenarios
2. Batch Processing: Group similar tasks (all upscaling together, all color grading together)
3. Parallel Generation: Run multiple tools simultaneously
4. Preset Automation: Create Photoshop actions/Lightroom presets for repetitive tasks
5. Quality Gates: Implement automated checks before proceeding to next phase

Common Pitfalls & Solutions

Pitfall	Impact	Prevention	Solution
Identity Drift	Model looks different across images	Strong reference embedding	Re-anchor to model sheet
Uncanny Valley	Viewer discomfort	Natural lighting, subtle imperfections	Add micro-details, reduce symmetry
Hand Issues	Unprofessional output	Pose selection, occlusion	Manual inpainting, ControlNet
Over-processing	Plastic/fake appearance	Light touch on enhancement	Reset and use minimal processing
Inconsistent Lighting	Disjointed portfolio	Document light setups	Standardize lighting presets

Premium Add-On Services

```
## Value-Add Services

### Motion & Animation
- [ ] Talking head video generation (D-ID, HeyGen)
- [ ] Body motion capture application
- [ ] GIF/Cinemagraph creation

### Customization Packages
- [ ] Aging/de-aging variants
```

```
- [ ] Seasonal appearance changes
- [ ] Special occasion styling

### Extended Assets
- [ ] Product integration photography
- [ ] Group/multi-avatar compositions
- [ ] Environmental storytelling series

### Technical Deliverables
- [ ] 3D model conversion (for VR/AR)
- [ ] Green screen/transparent backgrounds
- [ ] Layered PSD source files
```

📁 Client Delivery Template

Delivery Checklist Email

```
Subject: [Project Name] - AI Avatar Delivery Complete

Dear [Client Name],

Your AI avatar project is complete! Please find attached/link
ed:

📦 Delivery Package Contents:

1. Model Sheet (12 images)
   - 4 primary views (front, back, left, right profiles)
   - 8 secondary views (three-quarter angles, close-ups)

2. Expression Library (8 expressions)
   - Neutral through energetic range

3. Wardrobe Collection ([X] outfits)
```

```
   - Casual: [X] | Business: [X] | Formal: [X]

4. Contextual Imagery ([X] images)
   - Organized by scenario/environment

5. Web-Optimized Versions
   - Social media crops included
   - Multiple resolution options

📋 Technical Specifications:
- Primary resolution: [X]×[X] pixels
- Color space: [sRGB/Adobe RGB]
- File formats: [PNG/JPEG/TIFF]

🔐 Usage Rights:
[Insert licensing terms]

📞 Next Steps:
[Insert revision/feedback process]

Best regards,
[Your Name]
```

🔄 Continuous Improvement Framework

Version Control

- Maintain avatar versions (v1.0, v1.1, v2.0)
- Document all prompt refinements
- Archive successful generation parameters

Feedback Integration

- Collect client satisfaction scores
- Track common revision requests

- Update templates based on patterns

Technology Updates

- Monthly tool evaluation
- Quarterly workflow optimization
- Annual full process review

📚 Appendix

A. Prompt Library Templates

<details>
<summary>Click to expand full prompt library</summary>

Fashion/Editorial

```
[IDENTITY_TOKEN], high fashion editorial portrait, wearing [OUTFIT],
[POSE] pose, [ENVIRONMENT] setting, dramatic [LIGHTING] lighting,
shot on Phase One IQ4 150MP, 110mm lens, f/2.8, magazine cover quality,
Vogue aesthetic, sophisticated color grading, sharp focus, 8K resolution
```

Commercial/E-commerce

```
[IDENTITY_TOKEN], professional product photography model,
wearing [PRODUCT/OUTFIT], clean [POSE], white cyclorama background,
soft diffused studio lighting from 45 degrees, even exposure,
shot on Canon EOS R5, 85mm lens, f/8, e-commerce standard,
neutral color accurate, full body visible, centered composition
```

Lifestyle/Social

```
[IDENTITY_TOKEN], authentic lifestyle moment, [ACTION/ACTIVITY],
in [REAL_WORLD_LOCATION], [TIME_OF_DAY] natural lighting,
candid [EXPRESSION] expression, [OUTFIT] outfit,
shot on Sony A7IV, 35mm lens, f/2.0, shallow depth of field,
warm inviting color grade, Instagram aesthetic, genuine emotion
```

</details>

B. Tool Configuration Presets

<details>
<summary>Click to expand tool presets</summary>

Midjourney V6

```
--ar 3:4 --style raw --s 250 --c 15 --q 2
```

Stable Diffusion XL

```
Steps: 40
Sampler: DPM++ 2M Karras
CFG Scale: 7.5
Denoising Strength: 0.75
```

ControlNet Settings

```
OpenPose: Weight 1.0, Start 0, End 1
Canny: Weight 0.6, Start 0, End 0.8
IP-Adapter: Weight 0.85
```

</details>

C. Troubleshooting Guide

Issue	Likely Cause	Quick Fix
Blurry output	Low CFG, over-denoising	Increase CFG to 8-9, reduce denoise
Wrong ethnicity	Prompt ambiguity	Add explicit ethnicity descriptors
Age inconsistency	Missing age anchors	Include specific age in every prompt
Lighting mismatch	Context override	Add lighting keywords with emphasis
Pose distortion	Complex pose request	Use ControlNet pose reference